

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q19-1252

Fair Oaks 138 kV
February 15, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1252. The Interconnection Customer proposes a 91.9 MW battery energy storage facility interconnecting at the Fair Oaks 138 kV substation in the Pennsylvania area. The request was received on July 18, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 386 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$3,559,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Youngwood 500 kV/345 kV XFMR	Youngwood / Univ Of Nc Hapel Hill Cogen Facility 345.0 kV Line	110.5	732	6.4
Fair Oaks 345 kV/138 kV Autotransformer	East Devon / Carrel 345.0 kV	104.6	821	80.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Youngwood 500/345kV XFMR	Replace transformer with higher-rated unit	\$31,924,000	3% (\$981,000)
Fair Oaks 345-138 kV TX	Replace transformer with higher-rated unit	\$25,293,000	10% (\$2,578,000)
	Total allocated to project		\$3,559,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.